MIL-C-80111C
14 January 1987
SUPERSEDING
MIL-C-80111B
24 November 1975

MILITARY SPECIFICATION

CUTTING MACHINE, SHAPE, OXYGEN AND FUEL-GAS, ELECTRIC-MOTOR DRIVEN, TRACER GUIDED, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, electric-motor driven shape cutting machines which are capable of cutting identical shapes using a steel template to guide the cutting torch.

1.2 Classification. The cutting machine shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size 1 - Cutting capacity of not greater than a 4 inch to not less than a 19 inch diameter circle.

Size 2 - Cutting capacity of not greater than a 4 inch to not less than a 42 inch diameter circle.

Size 3 - Cutting capacity of not greater than a 4 inch to not less than a 72 inch diameter circle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433
DISTRIBUTION STATEMENT A. Approved for public release;distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/RMA/CGA IP-7 - Specification for Rubber Welding Hose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA PAMPHLET E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment.

(Application for copies should be addressed to the Compressed Gas Association, Inc., 500 Fifth Avenue, New York, NY 10110-0110.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The cutting machine shall be new and one of the manufacturer's current models capable of cutting shapes, circles, and straight lines in metal plates using an oxygen and fuel gas torch. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine . When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimension, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. All safety devices normally incorporated into the machine by the manufacturer for the protection of the operator shall be furnished. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all the requirements of CFR 29, CH XVII, Part 1910 that are applicable to the cutting machine itself. Additional safety and health requirements shall be as specified see (6.2.1).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible and identified.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components.

3.4.1 Frame assembly. The frame assembly shall be constructed with the required strength and rigidity to support and maintain alignment of all components necessary to provide the performance specified in 3.5.

3.4.2 Torch drive mechanism. The torch drive mechanism shall be electric motor driven and shall be of sufficient strength and rigidity necessary to perform as specified in 3.5. Sealed, permanently lubricated roller or ball bearings shall be provided at all rotating points.

3.4.3 Torch mounting mechanism. The torch mounting mechanism shall provide for mounting and positioning the torch in the vertical position. The torch positioning mechanism shall be continuously variable and shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

equipped with a positive locking device. The torch shall have sufficient vertical adjustment to accommodate the cutting capacity specified in 3.5. Vertical positioning shall be accomplished by rack and pinion gears. The arm that attaches to the frame shall be of sufficient strength to provide the accuracy specified in 3.5 with the torch in any position.

3.4.4 Cutting torch. Unless otherwise specified (see 6.2.1), the cutting machine shall be furnished with a cutting torch. The torch shall be of the machine type with a cylindrical barrel, a rack for vertical adjustment, gas valves, and a cutting tip. The torch shall be of the two or three hose style and unless otherwise specified (see 6.2.1), designed for use with oxygen and acetylene gas. The end of the torch housing and the fuel and oxygen valves shall be manufactured of bronze or brass. The torch shall be constructed to receive cutting tips of various sizes.

3.4.4.1 Cutting tips. Cutting tips shall be fabricated of copper or copper-tellurium and shall have a heat conductivity of not less than 90 percent of pure electrolytic copper. The tips shall have not less than four preheat holes and one cutting oxygen hole. The tips shall be permanently and legibly identified for size according to the manufacturer's standard commercial practice. The cutting tips shall provide a tight seal to prevent gas leaks around the tip nut. The center cutting oxygen orifice shall provide a stream of oxygen which travels down the centerline of the preheat flame group and shall not deviate from the center line by more than 2 degrees. A tip or set of tips capable of cutting SAE 1010 steel from 1/16 inch to four inches thick for the size 1 and 1/16 inch to seven inches thick for the size 2 and 3 shall be furnished. Additional tips shall be as specified (see 6.2.1).

3.4.5 Hoses and fittings. The cutting machine shall be furnished with all hoses and fittings required for operation. All hoses furnished shall comply with ANSI/RMA/CGA IP-7. The fittings attached to the hoses shall conform to CGA Pamphlet E-1. The hose fittings shall provide sufficient sealing to prevent gas leakage at all operating pressures. Unless otherwise specified (see 6.2.1), the machine shall be furnished with hoses not less than 25 feet in length.

3.4.6 Controls. All controls necessary for complete operation of the cutting machine functions shall be provided. The controls shall be located in a position convenient to the operator. The controls shall provide start, stop, and continuously variable speed control of the torch.

3.4.7 Magnetic tracer. The machine shall be furnished with a magnetic tracer designed to follow the edge of a steel template. The magnetic force holding the tracer to the template shall be sufficient to assure conformance with the accuracy specified in 3.5.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Unless otherwise specified (see 6.2.1), the machine shall be designed to operate from a 120 volt, 60 Hz, single phase power source.

3.4.9 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

3.5 Performance. The cutting machine shall be capable of cutting one half inch thick SAE 1010 steel at a speed of up to 25 inches per minute. The size 1 machine shall be capable of cutting SAE 1010 steel up to four inches thick and the size 2 and 3 up to seven inches thick at a speed of not less than three inches per minute. The size 1 machine shall be capable of cutting any shape that falls within a 19 inch diameter circle, the size 2 machine any shape that falls within a 42 inch diameter circle and the size 3 machine any shape that falls within a 72 inch diameter circle. Cutting accuracy shall be plus or minus 1/16 inch.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.7 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.9 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.9.1 Lubrication chart or plate. A lubrication chart or corrosion resistant plate shall be securely attached to each outfit. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following in formation:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humid-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

ity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The cutting machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be connected to the required electric power source and to an appropriate oxygen and fuel gas supply. All components of the gas system shall be checked for leaks. Gas pressures shall be set in accordance with the suppliers recommendations. The torch shall be lighted and adjusted to obtain a neutral flame. The machine shall be operated to insure conformance with all requirements specified herein without performing any cutting operations.

4.7.2. Performance tests.

4.7.2.1 Speed of cut test. A sample of SAE 1010 steel 1/2 inch thick by 25 inches long shall be cut. The time required to complete the cut shall be not greater than one minute. The cut edge of the kerf shall be square and the slag shall be easily removable.

4.7.2.2. Shape cutting test. A sample of SAE 1010 steel not less than 1/8 inch thick shall be cut. The manufacturers standard test template containing at least four direction changes shall be used to make the cut. The accuracy of the cut relative to the template shall be plus or minus 1/16 inch.

4.7.3. Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The cutting machine described in this specification is intended for cutting shapes, circles, and arcs from metal plate.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Cutting torch and fuel gas if different (see 3.4.4)

j. Additional cutting tips, if required (see 3.4.4.1)

k. Hoses and fitting, if different (see 3.4.5).

l. Electrical system and input power, if different (see 3.4.8).

m. Additional performance features, if required (see 3.5.1).

n. Standard equipment, if different (see 3.6).

o. Specify and fully describe optional equipment, if required (see 3.7).

p. Specify applicable repair parts, if required (see 3.8).

q. Nameplate, if different (see 3.9.1)

r. Hour meter, if different (see 3.10).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

s. First article inspection, if different (see 4.4).

t. Quality conformance inspection, if different (see 4.5).

u. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Cutting machine
Cutting machine flame
Flame cutting
Metal cutting
Oxygen-acetylene cutting machine

6.5 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

MIL-C-80111B	MIL-C-80111C
One type - One size	Size 1 (new size) Size 2 (new size) Size 3 (new size)

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(Project 3433-0108)

Review activities:
Navy - YD
Air Force - 84

User activity:
Army - AR

☆U.S. GOVERNMENT PRINTING OFFICE: 1987—704-036/70126

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-80111C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-23T04:59Z
Check the source to verify that this is the current version before use.